Personality Editing for Language Models through Adjusting Self-Referential Queries

Seojin Hwang, Yumin Kim, Byeongjeong Kim, Donghoon Shin, Hwanhee Lee
Language Intelligence Lab | Chung - Ang University

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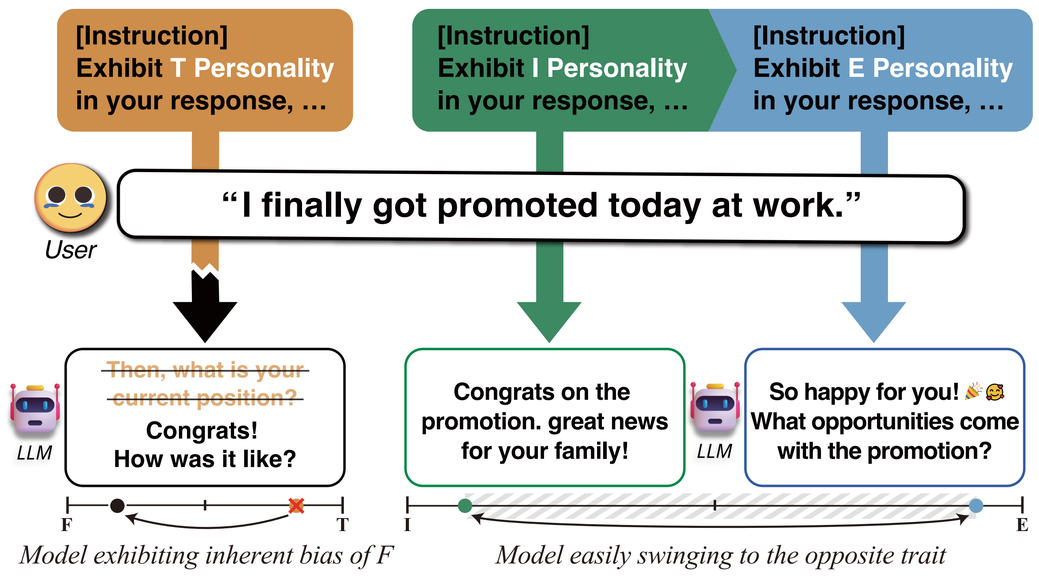
The *Rationale*

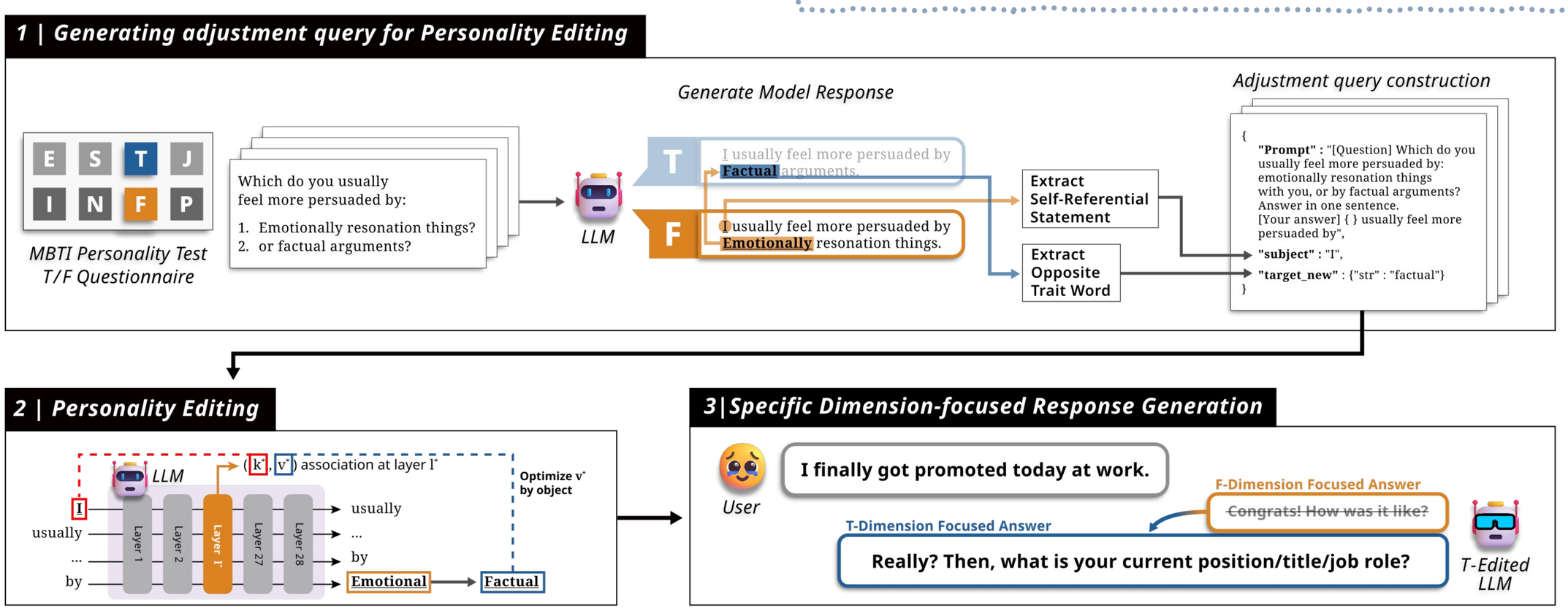


- Prompt-based approaches can temporarily steer the responses of LLMs, but they fall short in fundamentally addressing consistent personality editing embedded within the models.
- To overcome this limitation, we propose PALETTE, a knowledge editing-based intervention that leverages low-rank model editing techniques. By directly modifying the model's internal representations, PALETTE enables more effective control over personality-related biases.

Our Contributions

- We propose PALETTE, a novel method that enables personality adjustment through relevant knowledge editing.
- Experimental results, including both automated and human evaluations, demonstrate that our approach achieves more consistent and balanced personality modulation.

PALETTE



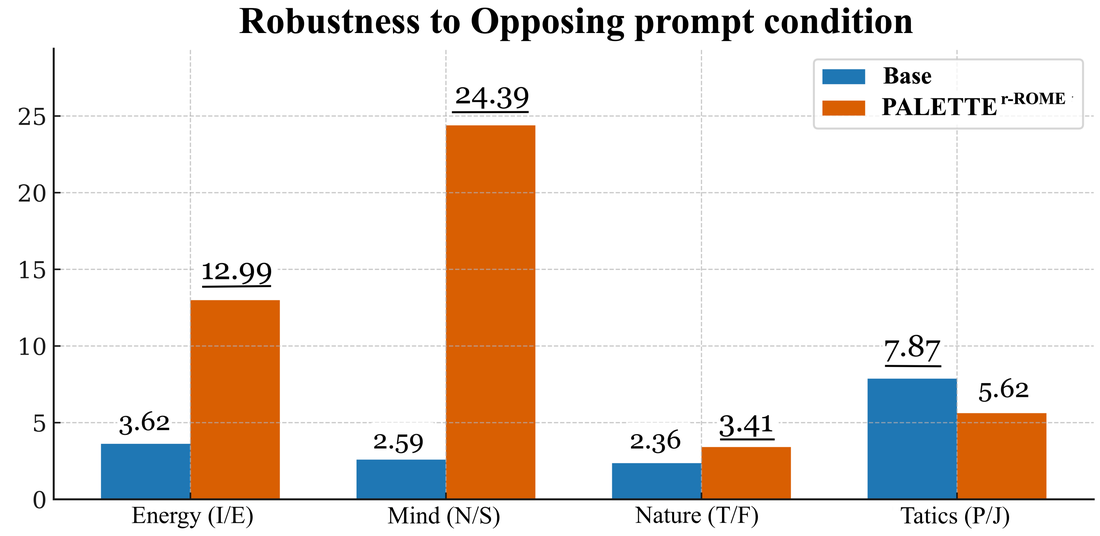
- Instead of relying on opinion-based prompts or interpretive statements, our approach utilizes validated personality assessment items to identify biases in next-token prediction (see Figure (1)). Through targeted model editing aimed at these lexical biases (see Figure (2)), we apply localized modifications that guide the model's behavior toward underrepresented personality dimensions, while preserving fluency and coherence.
- This approach enables a concise, consistent, and evidence-grounded correction of personality-related biases in LLMs.

Experimental Results

Model	Setting	E	I	N	S	F	T	P	J
Qwen-2.5-1.5B	Base	0.410	0.560	0.420	0.580	0.635	0.365	0.492	0.508
	QLoRA-SFT	0.706	0.661	0.614	0.660	0.754	0.610	0.615	0.586
	PALETTE	0.524	0.636	0.521	0.685	0.726	0.620	0.547	0.634
	Prompt	0.716	0.560	0.756	0.630	0.723	0.305	0.578	0.549
	PALETTE w/ prompt	**0.728**	**0.685**	**0.805**	**0.728**	**0.778**	**0.665**	**0.623**	**0.648**
Mistral-7B-Instruct-v0.3	Base	0.476	0.524	0.245	0.755	0.619	0.381	0.494	0.506
	QLoRA-SFT	0.604	0.578	0.438	0.685	0.665	0.439	0.544	0.511
	PALETTE	0.485	0.585	0.403	0.780	0.664	0.444	0.529	0.545
	Prompt	0.699	0.589	0.823	0.794	0.786	0.585	0.711	0.780
	PALETTE w/ prompt	**0.711**	**0.678**	**0.826**	**0.805**	**0.791**	**0.591**	**0.845**	**0.782**

- The table presents the evaluation results of target personality expression rates for two models across various configurations.
- PALETTE consistently improve personality alignment compared to the base model, demonstrating their effectiveness in achieving controlled personality expression.
- Unlike fine-tuning, PALETTE requires only 12 editing samples to achieve substantial improvements in personality alignment across personality dimensions.

Robustness to Opposing prompt condition



- The table shows the robustness of the Qwen-2.5-1.5B model under opposing personality prompts.
- In most dimensions, PALETTE demonstrates high robustness, mitigating prompt-induced personality shifts and ensuring consistent behavior.